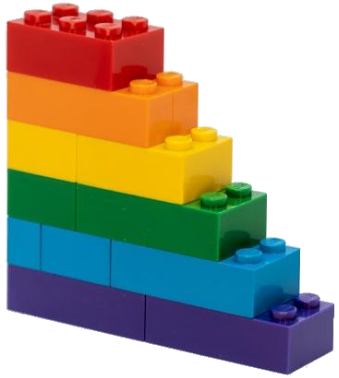
House Keeping

- Last session before the summer holidays! 🏖️ 😎 🌞
- Next session on 15th of September 2025. 🏎️

Azure Meetup
KONSTANZ



Reusability in IaC

The “**Art**” 🖼️ 🎨 🖌️ of **Module** development in Terraform

(by Stefan Rapp, 28th July 2025, 17:00 – 18:00)

HashiCorp Terraform

```
resource "azurerm_role_assignment" "before" {
  for_each                                = var.role_assignments_before != null ? var.role_assignments_before : {
  name                                    = each.key
  description                             = each.value.description
  scope                                   = try(var.role_assignment_scope_ids[each.value.scope_name], each.value.
  ole_definition_id                       = each.value.role_definition_id
  le_definition_name                      = each.value.role_definition_name
  ncipal_id                               = try(local.role_assignment_possible_principal_ids[each.value.principal_
  cipal_type                              = each.value.principal_type
  ition                                   = each.value.condition
  tion_version                            = each.value.condition_version
  ted_managed_identity_resource_id        = each.value.delegated_managed_identity_resource_id
  vice_principal_aad_check                = each.value.skip_service_principal_aad_check
}

nths ago | 1 author
erm_machine_learning_workspace" "this" {
                                          = [azurerm_role_assignment.before]
                                          = var.create_machine_learning_workspace ? 1 : 0
                                          = var.machine_learning_workspace_name
  e                                       = try(azurerm_resource_group.this[0].name, var.resource_group_name)
                                          = var.module_location
  id                                      = var.machine_learning_workspace_application_insights_id
                                          = var.machine_learning_workspace_key_vault_id
                                          = var.machine_learning_workspace_storage_account_id
                                          = var.machine_learning_workspace_container_registry_id
                                          = var.machine_learning_workspace_public_network_access_enabled
                                          var.machine_learning_workspace_image_build_compute_name
                                          ar.machine_learning_workspace_description
                                          machine_learning_workspace_friendly_name
                                          chine_learning_workspace_high_business_impact
                                          l.machine_learning_workspace_possible_identity_ids[var.machine_learni
                                          learning_workspace_v1_legacy_mode_enabled
                                          earning_workspace_sku_name

                                                                != null ? var.machine_learning_workspace_encryptions :

                                                                .key_vault_id, encryption.value.key_vault_id
                                                                d, encryption.value.key_id != null ? encrypt
                                                                ossible_identity_ids[encryption.value.use
```

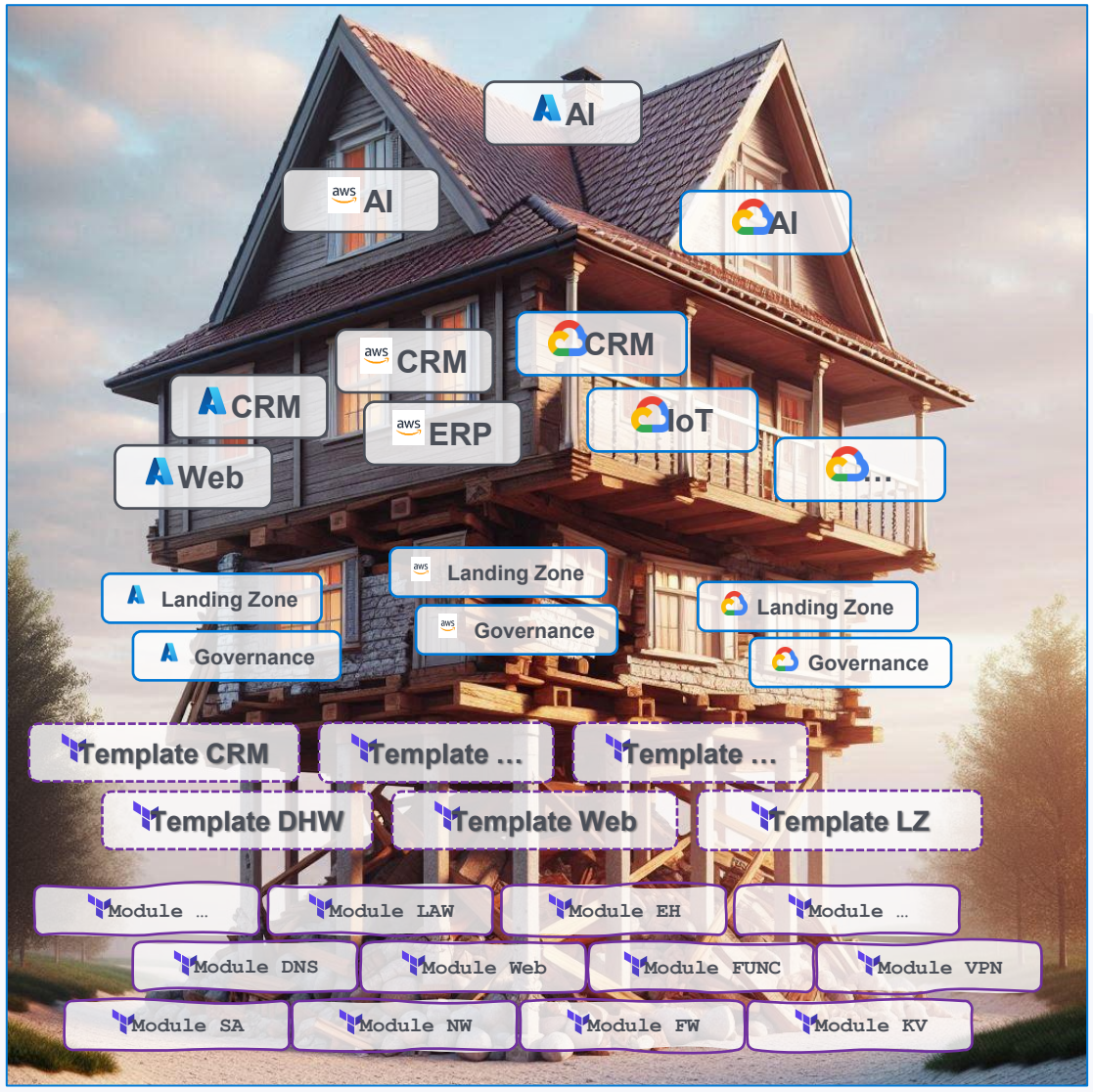
Table of contents

Why is modularization so essential?

From plane `#code` ➡️ to **modules**! 📦

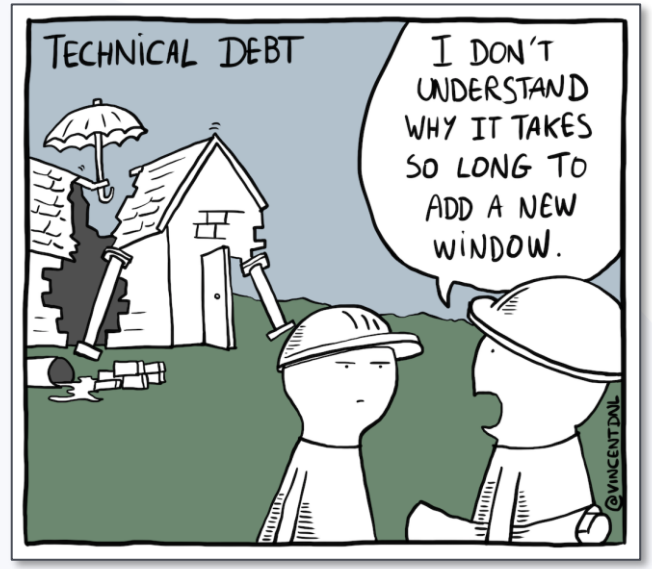
Baseline Knowledge

Why Modules are essential in IaC?



Consequences of bad modules?

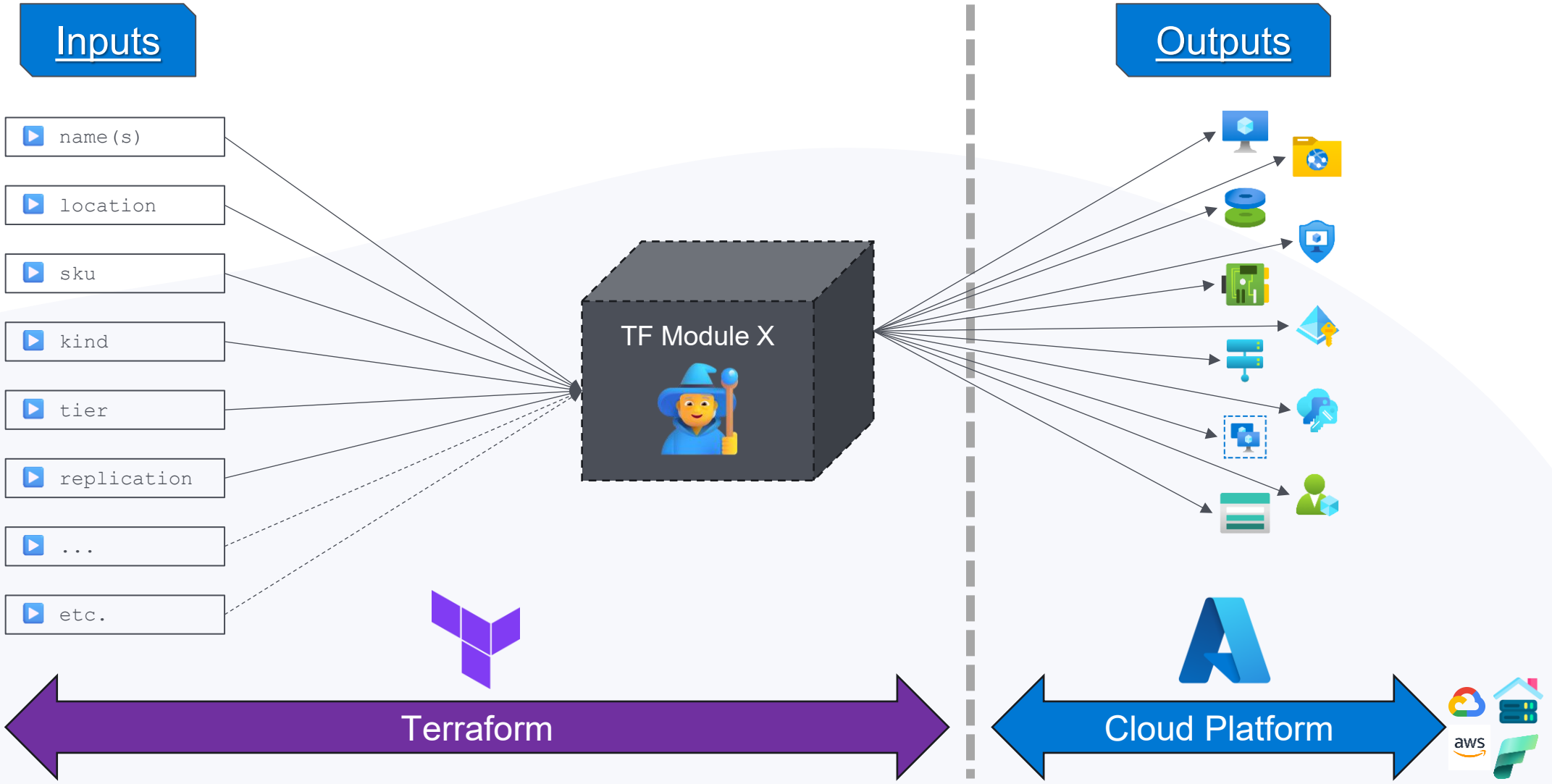
➡️ After the first change:



What does **modularization** mean Terraform?

Using Modules in Terraform

Accepts inputs & produces outputs



A module in Terraform is…

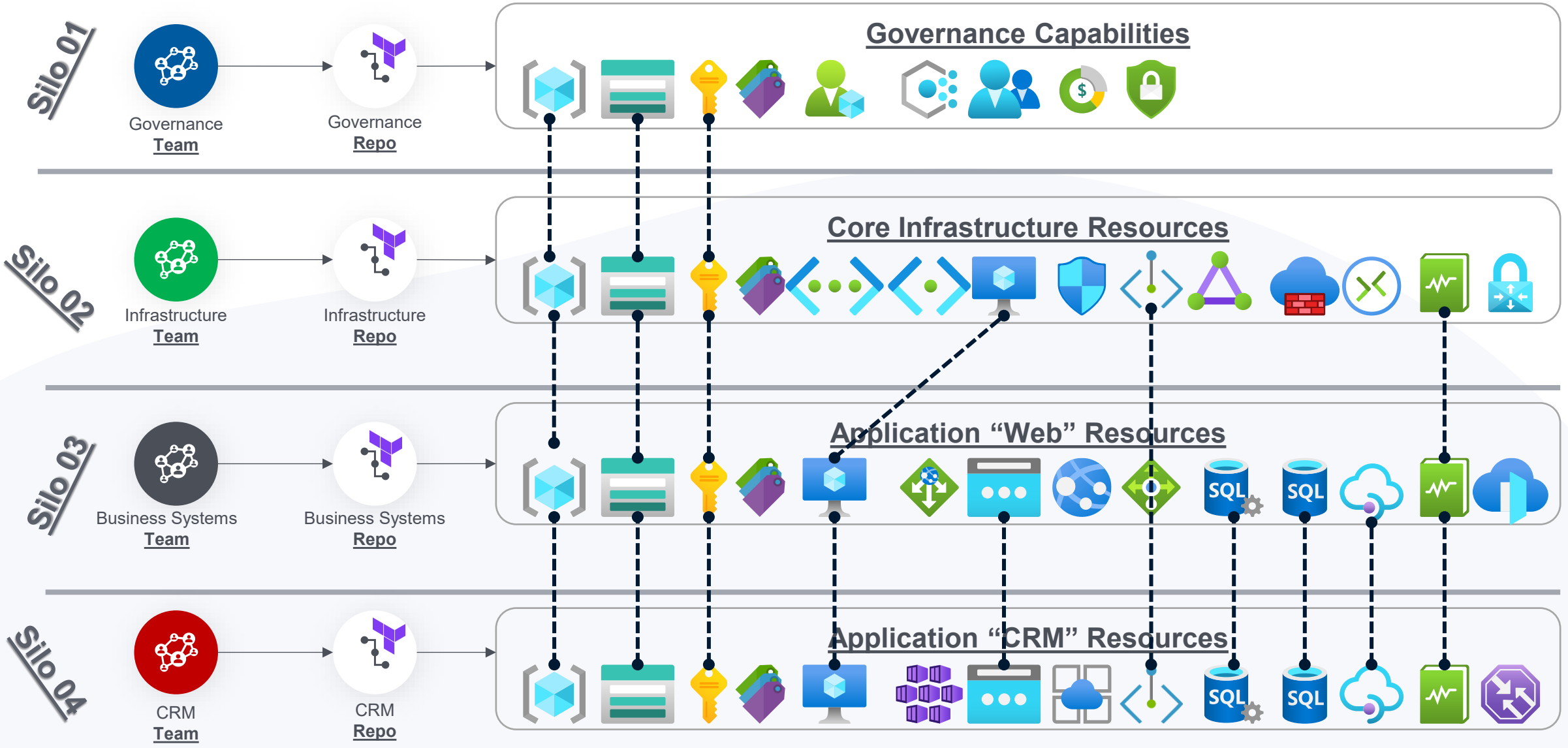
- the **definition** of infrastructure (“*What*?”). 🏛️
- independently **usable** & **testable**. 🛸
- **reusable** across different **projects** or **environments**. 🏝️
- **separated** in code repositories. 🪜
- used to **encapsulate** “complex” logic. 🤷
- is **easier** to manage, test, update and consume. 🔁

“A module is a container for multiple resources that are used together.”

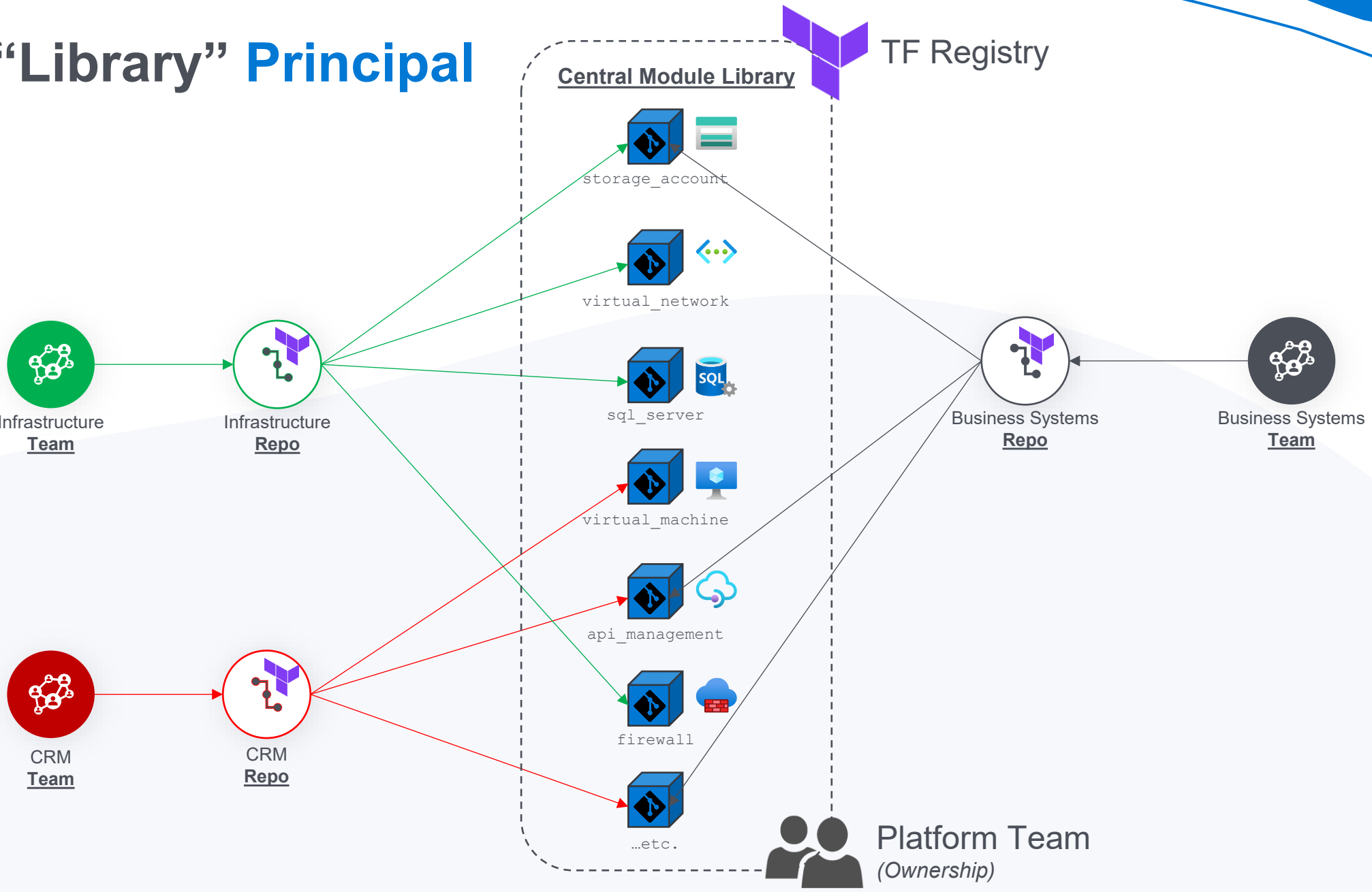
“A collection of Terraform files and folder in a repo.”

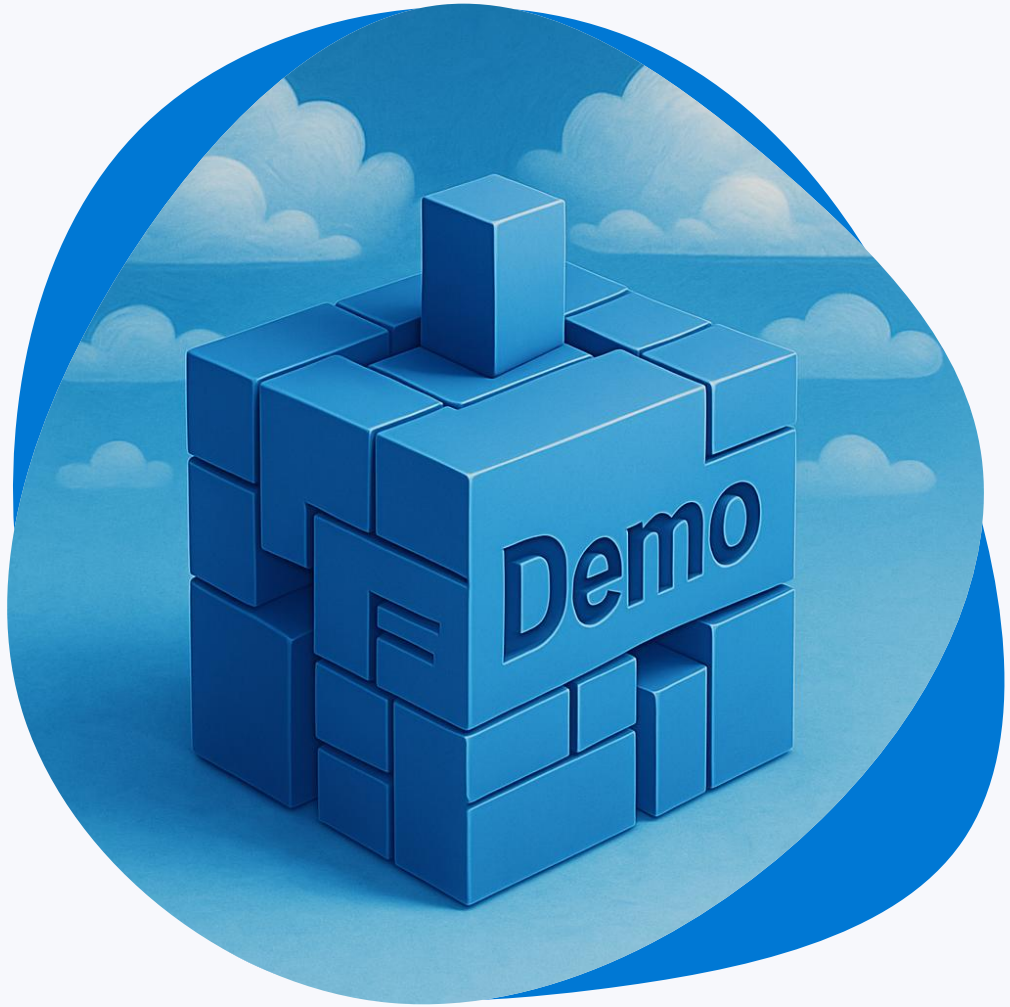
“Is the baseline for producing quality and resuable Terraform code.”

Current Situation in organization



The “Library” Principal





How does a **Module** look like in #Code?

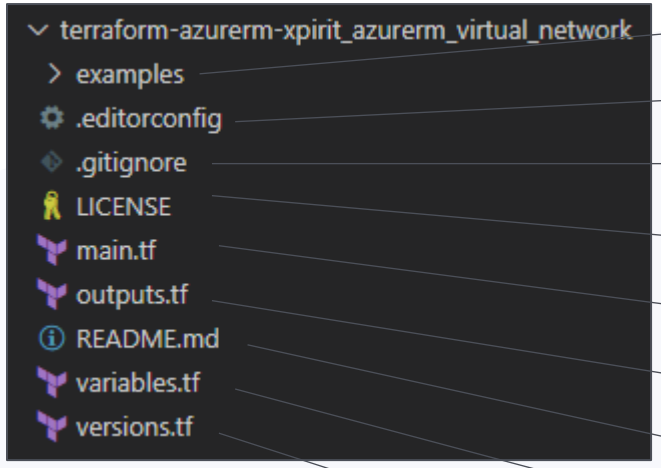
DEMO Time!

Standardized Module Structure 📦

1. **file** & **directory** layout

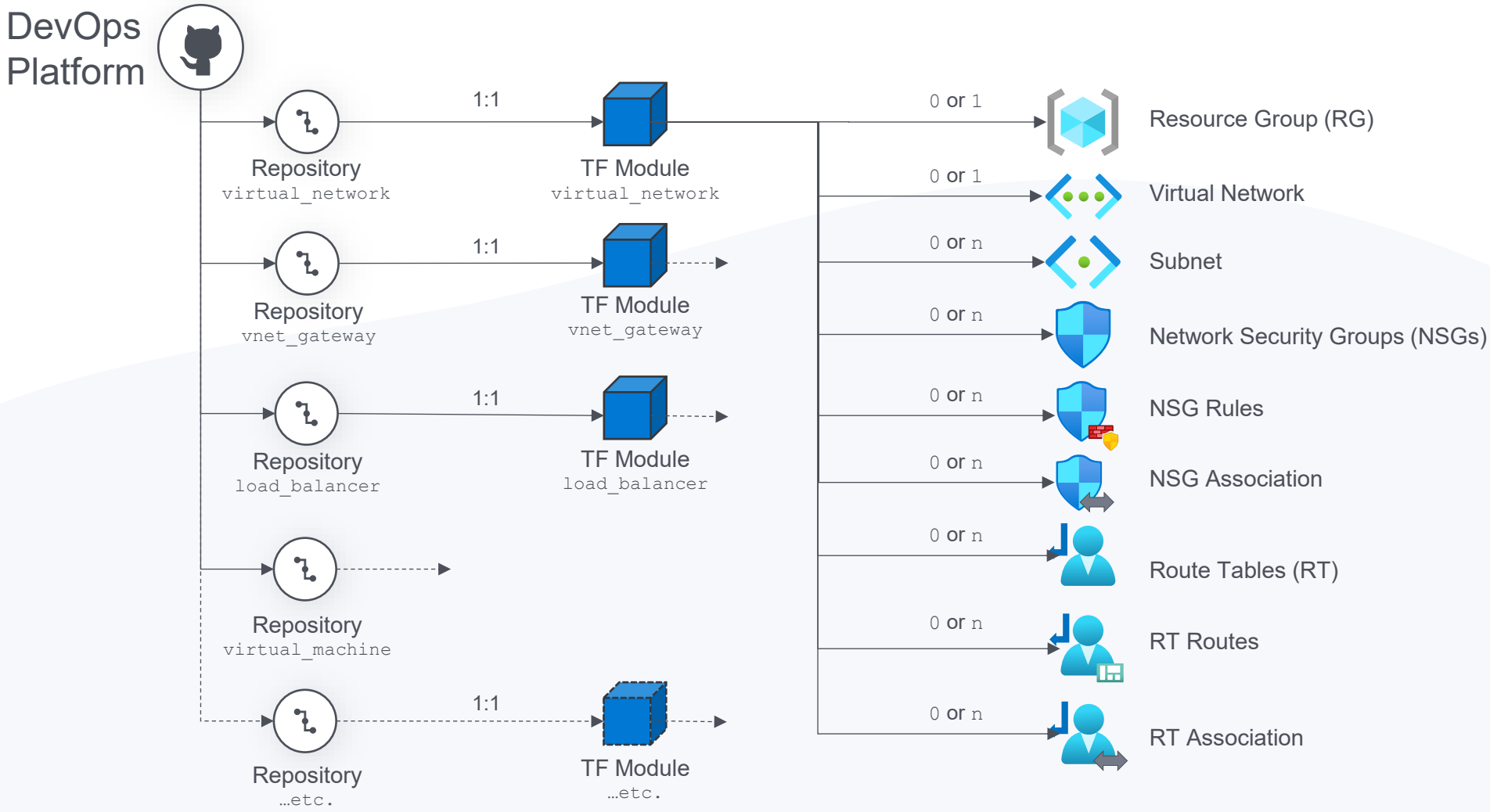
2. Distributed in separate repositories

3. Structure to generate documentation



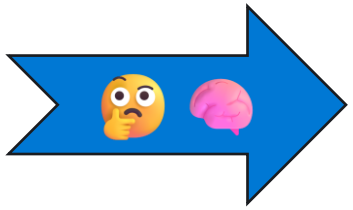
- `examples/` – separate subdirectory for each example incl. `README.md`
- `.editorconfig` – maintain consistent code style (Example)
- `.gitignore` – specifies untracked files that Git should ignore (Example)
- `LICENSE` – which this module is available (publicly, privately)
- `main.tf` – primary entry point, containing all needed resources
- `outputs.tf` – to return results to the calling module
- `README.md` – what the module should be used for (basic documentation)
- `variables.tf` – (declaration) to accept values from the calling module.
- `versions.tf` – (constraint) define supported terraform & provider version

Standard Module Architecture



Ask yourself… ⁉️

#1 Question



#2 Question

“What characterizes a…

- ***Virtual Network**?”*
- ***Storage Account**?”*
- ***Virtual Machine**?”*
- *…*
- *etc.*

*“How should the module look like so it can be used by **several** organizations, BUs, departments or workloads?“*

The “triple knowledge gap” problem

*“Writing modules is an **easy**, **dull** and **repetitive** task!* 😴

*Let our **trainees** do the job!”* 🤦

How to design reusable Modules? 🎨
- Structure?
- Patterns & Best Practices?
- Breaking Changes
- Lifecycle handling?

TF Module
📦 #code

TF Provider 🛠️

Cloud Platform ☁️

How Terraform sees the world? 👀
- Attributes?
- Dependencies?
- Version updates?
- Resource Behavior?

How the Cloud works under the hood? 🚙
- Resource configurations?
- Interdependencies?
- Limits, Quotas, Hidden Behaviors?
- Policies?

Use **Azure Verified Modules** (AVM) as Starting Point 🏎️

What are the “**Dos** & **Don’ts**“ regarding modules?

Best-Practices

Distinguish Bad from Good Modules

Bad Modules 😨

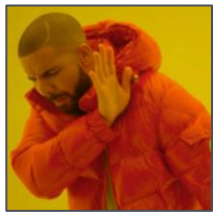
- … can drive you **nuts**. 🥜
- … are **horrible** to use. 👻
- … lead to bad **results**. 🤯

Good Modules 😌

- … are **fun** to use. 🥳
- … lead to **great** results. 👍
- … are **easy** to maintain and manage. 🪒

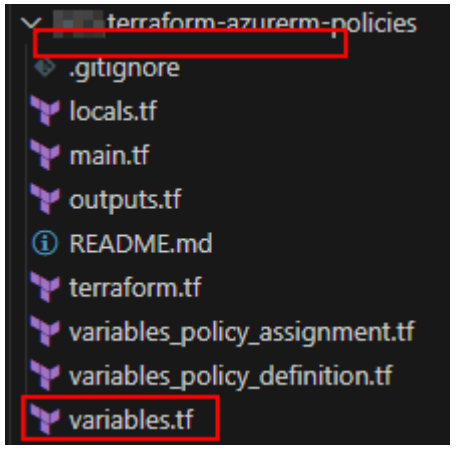
➔ Details: The **Art** of Modularization in Terraform – Mr. Azure 🖼️ 🎨 🖌️

Follow the “KISS” 👄 principal



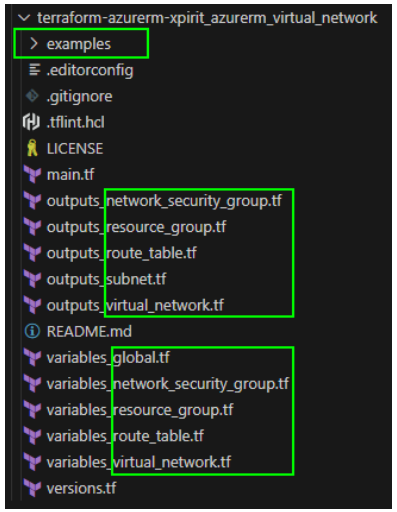
Bad Practice 👎

- Big monster 😈 repos with multiple modules 📁→📁→📁→…
- Missing `examples` folder
- Unstructured & hard to understand
- “Endless” `variables.tf` or `outputs.tf` files

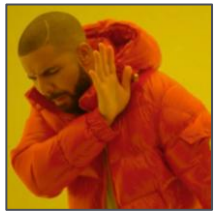


Good Practice 👍

- Each module is a separate **repo** → Define the **file** & **directory** layout.
- Philosophy → “*Doing one thing and doing it well*“. (single concern!)
- Small & self-contained 📦
- Serve a specific purpose → Discrete & specialized
- Independently testable 🔬 → `examples` folder
- Doing too much → ✂️ 🪓 Breaking it into smaller, more specialized modules.

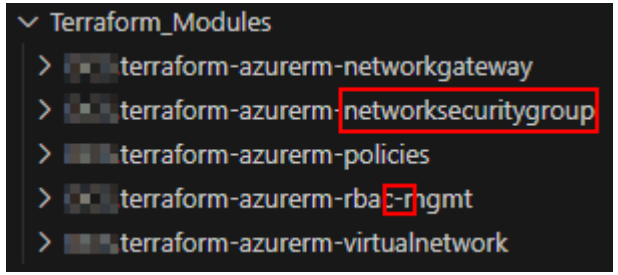


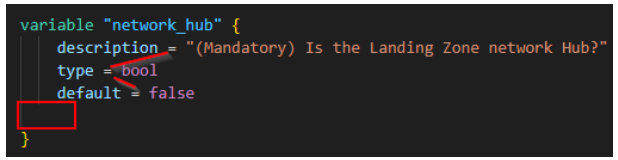
Repo & Content Structure

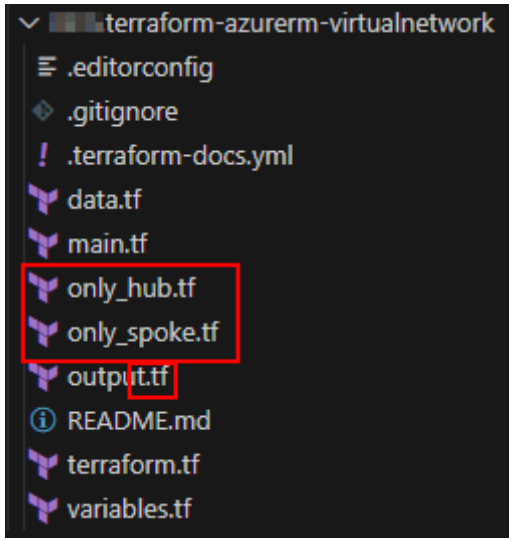


Bad Practice 👎

- No unique repo **naming**
- No standardized **structure** & **format**
- `data.tf` in modules
- No formatting & linting implemented

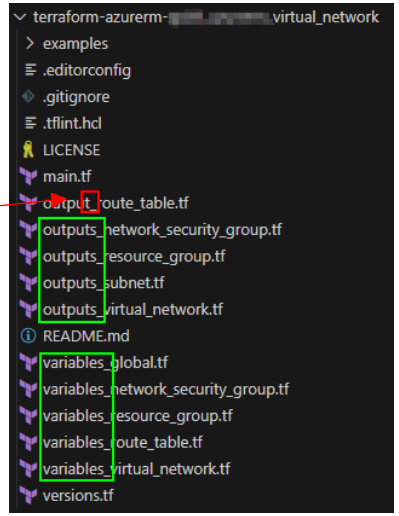




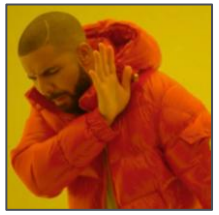


Good Practice 👍

- For orientation use Standard Module Structure (TF).
- Use a predefined repo **template**.
- Check, if all **files/folders** exist and **named** correctly. 😨
- Validate in the pipeline before the **merge**.
- Use that structure to generate **documentation** (`README.md`).
- Avoid **nested** modules, due to complexity.



Proper Naming Convention



Bad Practice 👎

- Each developer uses individual **names**.
- No standardized **abbreviations** or **prefixes**.
- Each module looks differently.
- Hard to troubleshoot.

```
resource "azurerm_subnet" "vpn-gw-subnet" {
  name                 = "GatewaySubnet"
  resource_group_name  = var.hub_resource_group_name
  virtual_network_name = var.hub_vnet_name
  address_prefixes     = [var.vpn_vnet_subnet_prefix]
}

# Associate gateway subnet to hub routing-tables
data "azurerm_route_table" "route-table-hub" {
  name                = var.hub_route_table_name
  resource_group_name = var.hub_resource_group_name
}

resource "azurerm_subnet_route_table_association" "rt_association" {
  subnet_id      = azurerm_subnet.vpn-gw-subnet.id
  route_table_id = data.azurerm_route_table.route-table-hub.id
}
```

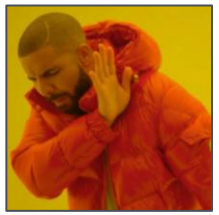
Good Practice 👍

- Define the standard before the **start**.
- Stick to it!
- Agree on **naming** conventions, **abbreviations** and **prefixes**.
- **Lint** the code before the merge.
- Strive for **consistent** and **descriptive** names.

```
resource "azurerm_virtual_network_peering" "this" {
  count                        = var.create_virtual_network_peering ? 1 : 0
  name                         = var.virtual_network_peering_name
  resource_group_name          = try(azurerm_resource_group.this[0].name, var.resource_group_name)
  virtual_network_name         = try(azurerm_virtual_network.this[0].name, var.virtual_network_peering_vir
  allow_forwarded_traffic      = var.virtual_network_peering_allow_forwarded_traffic
  allow_gateway_transit        = var.virtual_network_peering_allow_gateway_transit
  allow_virtual_network_access = var.virtual_network_peering_allow_virtual_network_access
  remote_virtual_network_id    = var.virtual_network_peering_remote_virtual_network_id
  triggers                     = var.virtual_network_peering_triggers
  use_remote_gateways          = var.virtual_network_peering_use_remote_gateways
}

resource "azurerm_subnet" "this" {
  for_each                          = var.subnets != null ? var.subnets : {}
  name                              = each.key
  resource_group_name               = try(azurerm_resource_group.this[0].name, var.resource_group_name)
  address_prefixes                  = each.value.address_prefixes
  private_endpoint_network_policies = each.value.private_endpoint_network_policies
  service_endpoints                 = each.value.service_endpoints
  virtual_network_name              = try(azurerm_virtual_network.this[0].name, var.virtual_network_name)
```

Do not hardcode 🪨 values



Bad Practice 👎

- Do not use fixed patterns
- Avoid hard-coded values
- Not changeable
- Unflexible

```
resource "azurerm_virtual_network_gateway" "vpn-gateway" {
  name                = "vgw-${var.location_code}-${var.name}-${var.environment}-${random_string.id.result}"
  location            = var.location
  resource_group_name = var.hub_resource_group_name

  type     = "Vpn"
  vpn_type = "RouteBased"

  active_active = false
  enable_bgp    = false
  sku           = "VpnGw1"


  ip_configuration {
    name                          = "vnetGatewayConfig"
    public_ip_address_id          = azurerm_public_ip.vpn_public_ip.id
    private_ip_address_allocation = "Dynamic"
    subnet_id                     = azurerm_subnet.vpn-gw-subnet.id
  }

}
```

Good Practice 👍

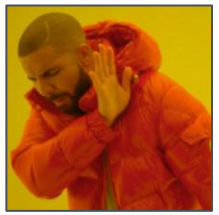
- Attributes mapped to a variable.
- Use `default` values in `variables.tf` file.
- `default` values are **overwritable**.

```
resource "azurerm_virtual_network_gateway" "this" {
  count                           = var.create_virtual_network_gateway ? 1 : 0
  location                        = var.module_location
  resource_group_name             = try(azurerm_resource_group.this[0].name, var.resource_group_name)
  name                            = var.virtual_network_gateway_name
  active_active                   = var.virtual_network_gateway_active_active
  default_local_network_gateway_id = try(azurerm_local_network_gateway.this[var.virtual_network_gateway
  enable_bgp                      = var.virtual_network_gateway_enable_bgp
```

```
variable "virtual_network_gateway_type" {
  description = "The type of the Virtual Network Gateway. Valid op
  type        = string
  default     = "Vpn"
}

variable "virtual_network_gateway_vpn_type" {
  description = "The routing type of the Virtual Network Gateway.
  type        = string
  default     = "RouteBased"
}
```

Keep modules “DRY” ☂️



Bad Practice 👎

- Code duplications (same logic → multiple places)
- Reduced readability (more lines of code)
- Higher maintenance effort
- Scalability issues (“enterprise scale”)

```
resource "random_password" "[redacted]-datacenter" {
  length           = 32
  special          = true
  override_special = "!#$%&*()-_=+[]{}<>:?"
}

resource "random_password" "[redacted]" {
  length           = 32
  special          = true
  override_special = "!#$%&*()-_=+[]{}<>:?"
}
```

Good Practice 👍

- Don’t repeat yourself (DRY)
- Do not use a `resource` multiple times within the same module.
- Use `for_each` loops or `count` in module resources. (Article)
- Try to use unique identifier `this` for the resource.

```
resource "azurerm_public_ip" "this" {
  for_each                = var.public_ips != null ? var.public_ips : {}
  location                = var.module_location
  resource_group_name     = try(azurerm_resource_group.this[0].name, var.
  name                    = each.key
  allocation_method       = each.value.allocation_method
  ddos_protection_mode    = each.value.ddos_protection_mode
  ddos_protection_plan_id = each.value.ddos_protection_plan_id
```

How to **manage** a module according to its lifecycle?

Publishing & Versioning

Use a Terraform Registry

TF Cloud helps teams to **provision** infrastructure.

Support for **versioning** and list of **providers** and **modules**.

Handles **downloads** and **access** with TF Cloud API tokens.

Consumers do not need **direct** access to the module's repository.

Public Registry

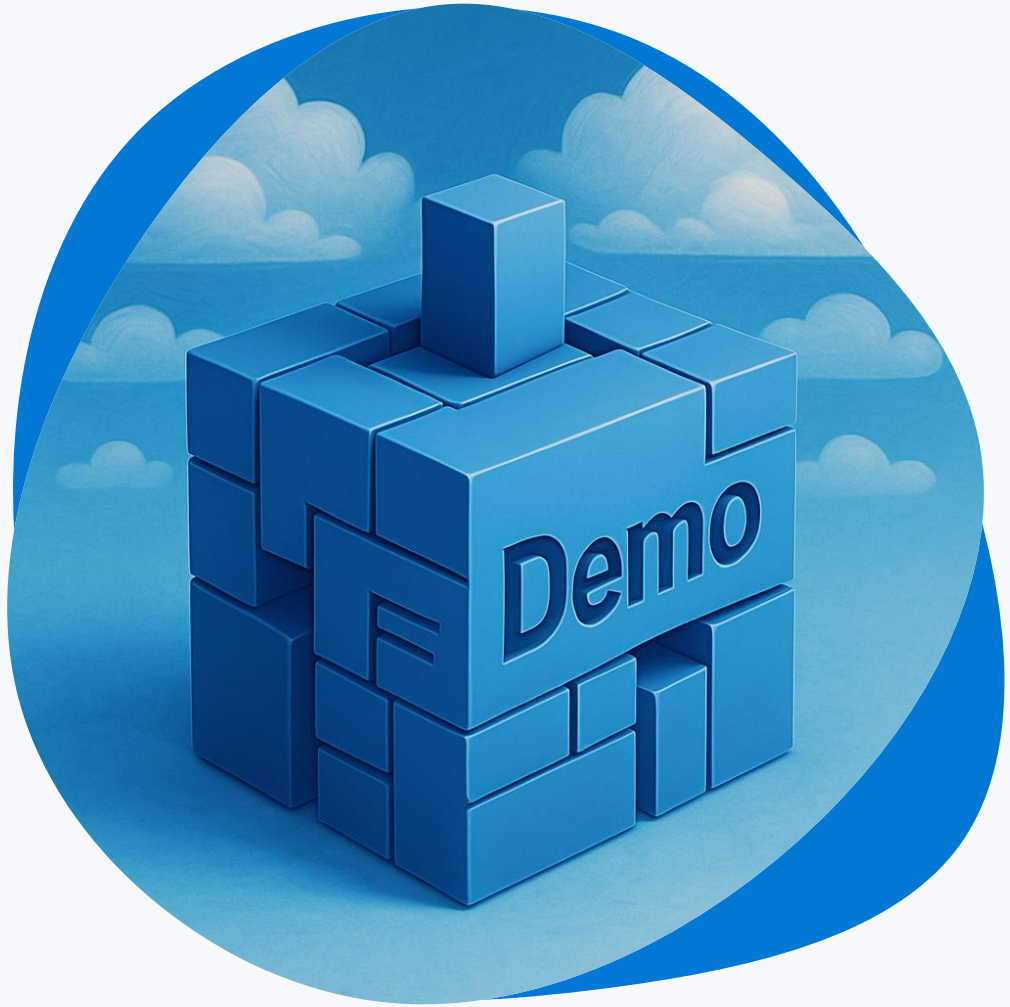
https://**registry**.terraform.io

Share Terraform providers and Terraform modules public.

Private Registry

https://**app**.terraform.io

Hosted on an organization's private registry and are only available to members

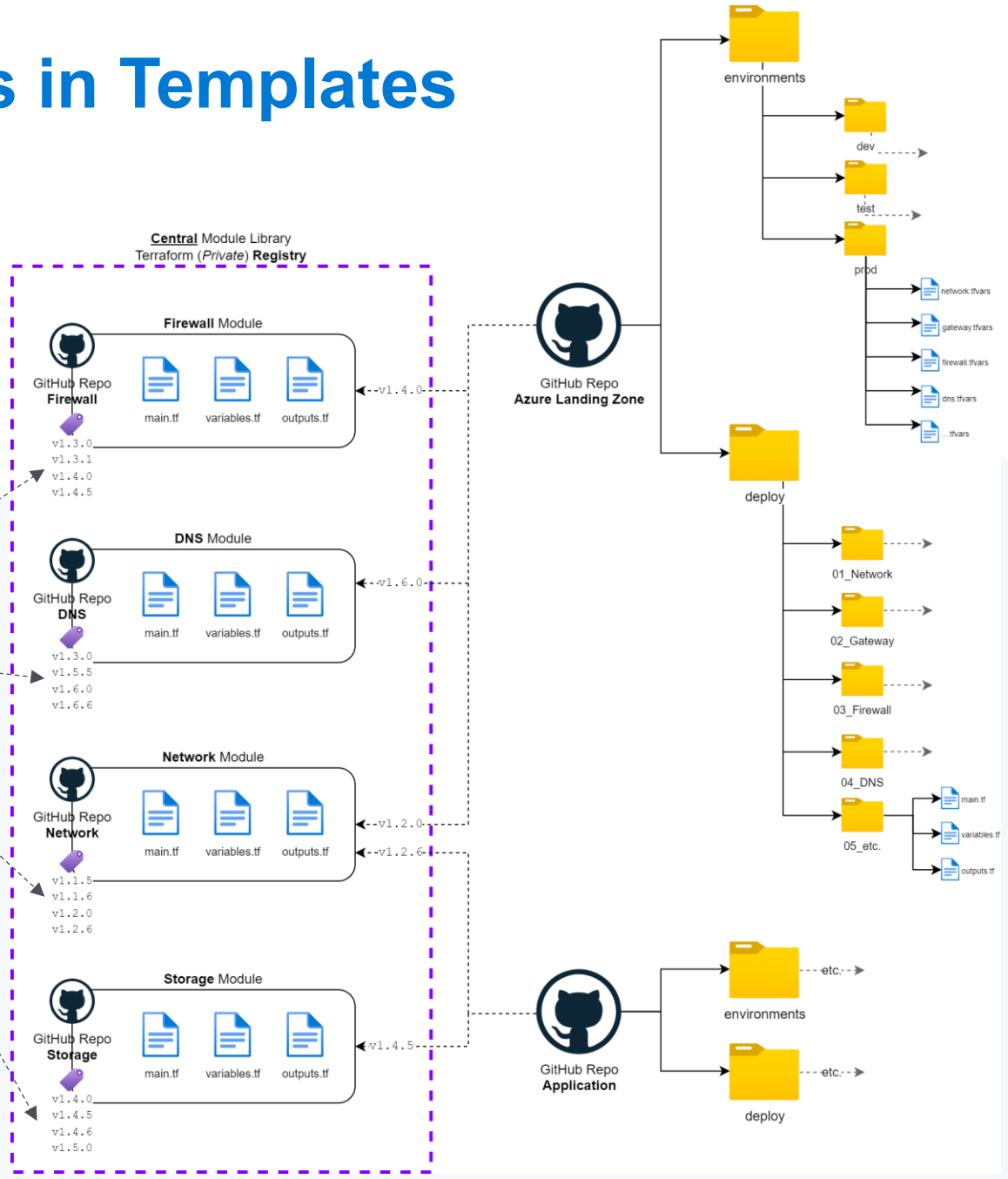


How does a **Registry** look like?

DEMO Time!

Reference Modules in Templates

Use SemVer:



How to use modules in Templates?

- Local file system

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {
  source = "../../../module-library-azurerm/terraform-azurerm-xpirit_azurerm_management_group"

  create_management_group                                = true
  management_group_name                                  = "c06835d9-1135-4b19-b403-749c06443f4c"
  management_group_display_name                          = "mg-pritix"
  management_group_subscription_association_subscription_ids = []
}
```

- Directly from the repository

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {
  source = "git::https://github.com/rapster83/terraform-azurerm-xpirit_azurerm_management_group?ref=v3.42.0-0.0.3"

  create_management_group                                = true
  management_group_name                                  = "c06835d9-1135-4b19-b403-749c06443f4c"
  management_group_display_name                          = "mg-pritix"
  management_group_subscription_association_subscription_ids = []
}
```

- Private registry

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {
  source  = "app.terraform.io/Customer-ML-Pritix/xpirit_azurerm_management_group/azurerm"
  version = "3.42.0-0.0.2"

  create_management_group                                = true
  management_group_name                                  = "c06835d9-1135-4b19-b403-749c06443f4c"
  management_group_display_name                          = "mg-pritix"
  management_group_subscription_association_subscription_ids = []
}
```

- Public registry (→ TF Registry)

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {
  source  = "azrappster83/management_group/azurerm"
  version = "4.0.0"

  create_management_group                                = true
  management_group_name                                  = "c06835d9-1135-4b19-b403-749c06443f4c"
  management_group_display_name                          = "mg-pritix"
  management_group_subscription_association_subscription_ids = []
}
```

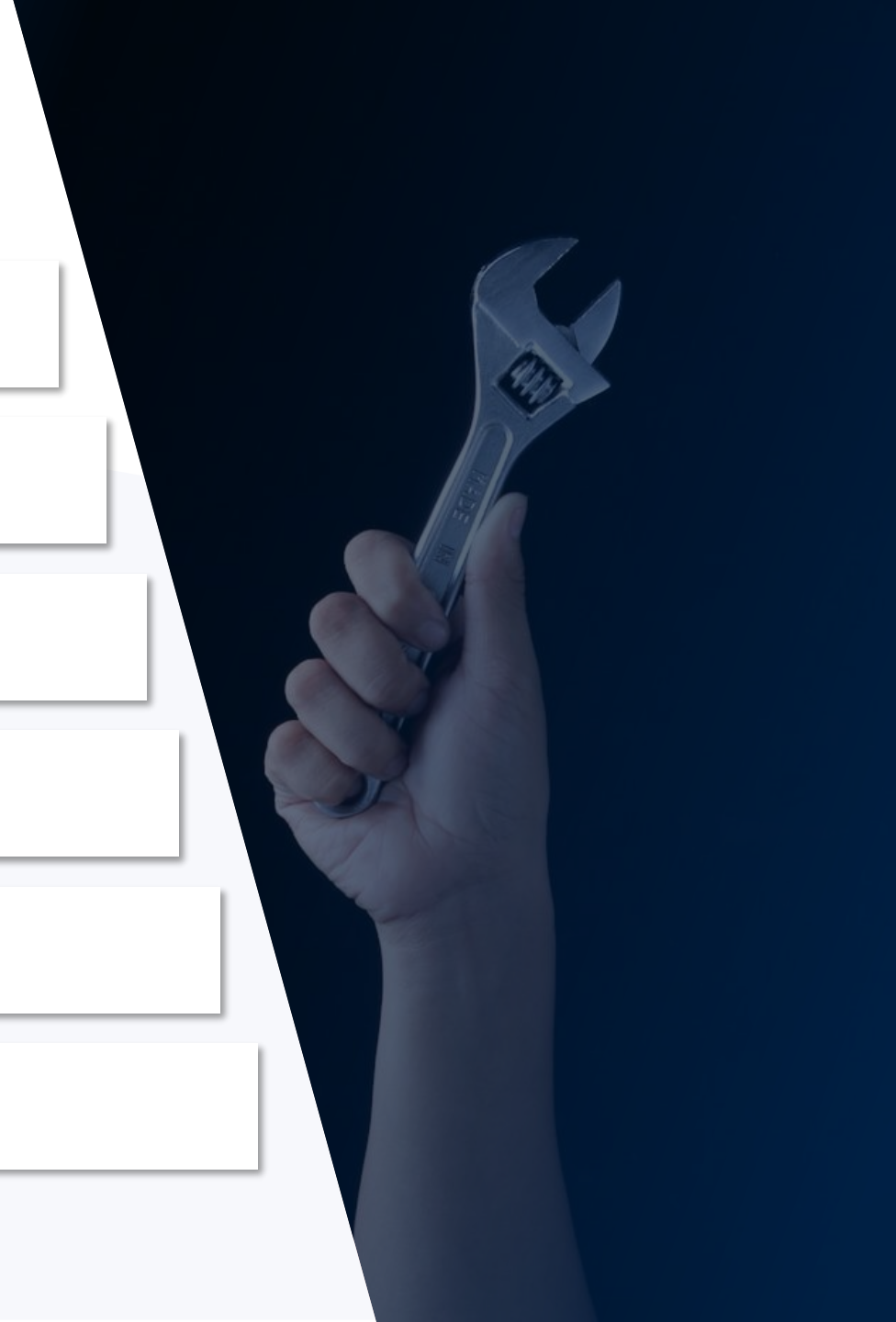
Use SemVer

Which **tools** should be used to improve module quality?

Additional Tooling

Tools for Terraform

- **tflint** → Linter for Terraform code + custom rules 📖
- **checkov** → Code analysis (security & compliance) 🔐
- **snyk** & **trivi** → Static code analysis 🔐
- **terraform-docs** → Doc from input/output definitions 📄
- **terratest** (Go) → Framework for writing automated tests 🧪
- **tfupdate** → Automates version updates 🔁



What are the **essentials** of the session?

Key Takeaways

🔑 Key Takeaways

- Start Modularization from the **beginning**! 🔝
- Keep modules **flexible** (“*Do not hardcode*”). 📏
- Follow the code **best-practices**. ✅
- Make sure that all modules “*look the same*” (**standardization**). 👁️
- Use a **central** module library across projects or departments. 🏛️
- Version your modules using **Tags**. 🏷️
- Use proper **Tools** to improve code quality. 🛠️

PROFILE – Speaker

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#AzureRocks 🤘 🤷‍♂️ 🎸

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